Worksheet 4 – March 6, 2026

Packet Detective Lab: HTTP vs HTTPS

🎯 **Objective:**

By the end of this worksheet, you will be able to observe unencrypted vs encrypted network traffic, investigate what data could be seen by an attacker, and explain why HTTPS protects sensitive information.

🔐 Part 1: Wireshark Scavenger hunt

📝 **Task:**

1. Start a new capture on Wireshark (mode any)
2. Then open Firefox and browse around to a few websites
3. See if you can find the following things in Wireshark:

(Hint: You may need to visit multiple different websites)

1. Find a DNS request

What website is being looked up?

Answer: _____

2. Find an IP Address

Write down one IP address you see in a packet.

Answer: _____

3. Find a packet that uses HTTP

What website is being visited?

Answer: _____

4. Find a packet that uses TLS or HTTPS

Why can't we read the website data in these packets?

Answer: _____

HTTP Investigation

5. Find a GET request

What page is the browser requesting?

Answer: _____

6. Find a POST request

What do you think the user might be doing when this request happens?

Example: logging in, submitting a form, etc.

Answer: _____

7. Find a Server Response

Look for a response like:

200 OK
404 Not Found

What response did you find?

Answer: _____

Packet Detective Challenge

8. Follow a TCP Stream

Right click a packet → **Follow → TCP Stream**

What conversation do you see?

Answer: _____

9. Find the User Agent

This tells you information about the browser.

Example: Chrome, Firefox, etc.

Answer: _____

10. Bonus Challenge

Can you find:

- a **DNS packet**
- an **HTTP packet**
- a **TLS packet**

Write the **packet numbers**.

DNS Packet #: _____
HTTP Packet #: _____
TLS Packet #: _____

🌐 Part 2: Investigate Wireshark

📝 **Task:**

1. Open Wireshark on your own computer.
2. Start a capture on local interface
3. Go to the terminal and navigate to the directory with server.py
4. Run python3 server.py in the terminal
5. Visit http://localhost:8000 on firefox
6. Enter fake credentials to the website

👀 Observe:

- What does the traffic look like?
- Can you see your username and password?

🌐 Part 3: Mystery PCAP Challenge

📝 **Task:**

- Load the pre-captured mystery.pcap provided by the instructor.
- Use the filters:

http
tls
dns

Your mission: Solve the detective challenges:

1. Find the first HTTP login (username/password visible)
2. Find the HTTPS login (content encrypted)
3. Identify which sites were visited (DNS queries)
4. Label each action as SAFE / EXPOSED / PARTIALLY EXPOSED

💬 **Discussion:**

- Which login was easiest to read?
- Which was impossible to read?
- What surprises you about the DNS visibility?